

5(a)(i)	(two) waves travelling (at same speed) in opposite directions overlap	B1
	waves (of the same type) have same frequency/wavelength	B1
5(a)(ii)	phase difference = 0	A1

PUBLISHED

Question	Answer	Marks
5(b)(i)	$f_0 = f_s v / (v - v_s)$ $543 = f \times 334 / (334 - 13)$	C1
	$f = 522 \text{ Hz}$	A1
5(b)(ii)	(the speed is) decreasing	B1
5(c)(i)	$I \propto A^2$	B1
	$I_T / I_0 = \cos^2 20^\circ$ or $A_T / A_0 = \cos 20^\circ$	C1
	ratio = 0.94	A1
5(c)(ii)	angle = 140°	A1

Question	Answer	Marks
----------	--------	-------